Communication Theory

Name: A. Karthik

Roll-No: 2023102009

(1) passband signals

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

(a) $u_p(t) = \text{Re}\{u(t)e^{j2\pi f_c t}\}$ — ①

$u(t)$ is complex envelope of $u_p(t)$.

Now,

$$u_p(t) = \text{sinc}(2t)\cos(100\pi t)$$

~~for it is same as~~

~~$u_p(t) = \text{sinc}(2t)e$~~ $e^{j2\pi(50)t} = \cos(100\pi t) + j\sin(100\pi t)$

$\therefore u_p(t) = \text{Re}\{\text{sinc}(2t)e^{j2\pi(50)t}\}$ — ②

$\therefore \boxed{u(t) = \text{sinc}(2t)} \Longrightarrow$ From comparing ① & ② eqn's.

Now, for $v_p(t)$,

$$v_p(t) = \text{sinc}(t)\sin(101\pi t + \pi/4)$$

$$= \text{sinc}(t)\cos(\pi/2 - (101\pi t + \pi/4))$$

$$= \text{sinc}(t)\cos(\pi/4 - 101\pi t)$$

we know $\cos(-\theta) = \cos\theta$)

$$\therefore = \text{sinc}(t)\cos(101\pi t - \pi/4)$$

$$\therefore v_p(t) = \text{Re}\{\text{sinc}(t)e^{j(101\pi t - \pi/4)}\}$$

$$= \text{Re}\{\text{sinc}(t)e^{-j\pi/4}.e^{j\pi t}.e^{j100\pi t}\} \quad \text{— ③}$$

By comparing ① & ③

$$\therefore v(t) = \text{sinc}(t)e^{j(\pi t - \pi/4)}$$

$$\therefore v(t) = \text{sinc}(t)\left[\cos(\pi t - \pi/4) + j\sin(\pi t - \pi/4)\right]$$

$\because \ v(t) = \text{sinc}(t)\left[\cos(\pi t - \pi/4) + j\sin(\pi t - \pi/4)\right]$

$\Downarrow$

$V(f) = I_{[-1/2, 1/2]} \cdot e^{-j\pi/4} * \delta(f - 1/2)$

$= I_{[0,1]} e^{-j\pi/4}$

(b) $u_p(t) = \text{sinc}(2t)\cos(100\pi t)$

We know,

$W \text{sinc}(Wt) \longleftrightarrow I(f)_{[-W/2, W/2]}(f)$

$\therefore$ so, $\text{sinc}(2t) \xleftrightarrow{FT} \frac{1}{2} I_{[-W/2, W/2]}(f)$ $\boxed{W = 2}$

$= \cancel{\frac{1}{2} I_{[-1/2, 1/2]}(f)}$

$= \frac{1}{2} I_{[-1,1]}(f)$

Also,

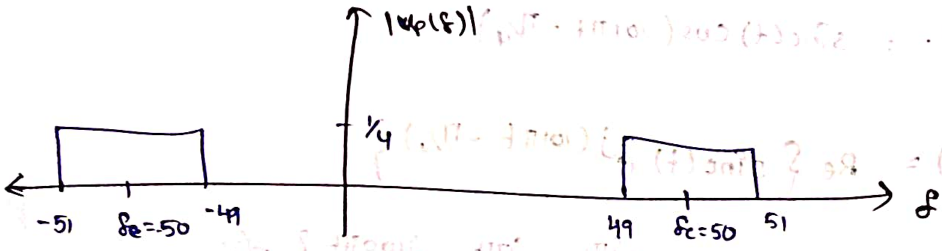
$m(t)\cos(2\pi f_c t) \longleftrightarrow \frac{1}{2}\left[M(f - f_c) + M(f + f_c)\right]$

$\therefore$ for Here We can take $\text{sinc}(2t)$ as $m(t)$

For $\text{sinc}(2t)\cos(100\pi t) \longleftrightarrow$ $\cancel{\frac{1}{4} I}$ $f_c = 50$

$\frac{1}{2}\left[\frac{1}{2} I_{[-1,1]}(f) * \left[\delta(f - f_c) + \delta(f + 50)\right]\right]$

$= \frac{1}{4}\left[I_{[49, 51]}(f) + I_{[-51, -49]}(f)\right]$



Between,

Bandwidth of $u_p(t) = 51 - 49 = 2\text{Hz}$

$\therefore$ Bandwidth of $u_p(t)$ $= 2\text{ Hz}$

$v_p(t) = \operatorname{sinc}(t)\sin(101\pi t + \pi/4)$

We know $\operatorname{sinc}(t) \longleftrightarrow I_{[-1/2, 1/2]}(f)$

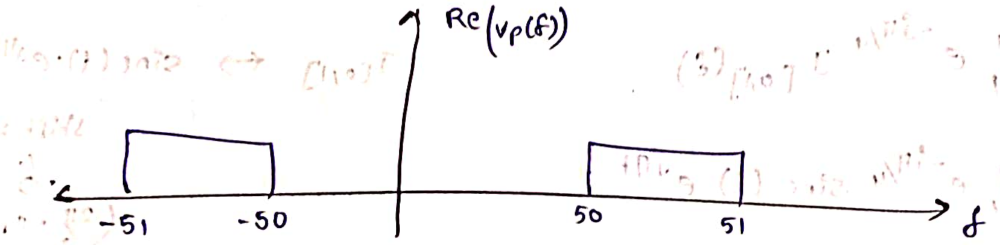
$$v_p(t) = \operatorname{sinc}(t)\left[\sin(101\pi t)\cdot\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\cos(101\pi t)\right]$$

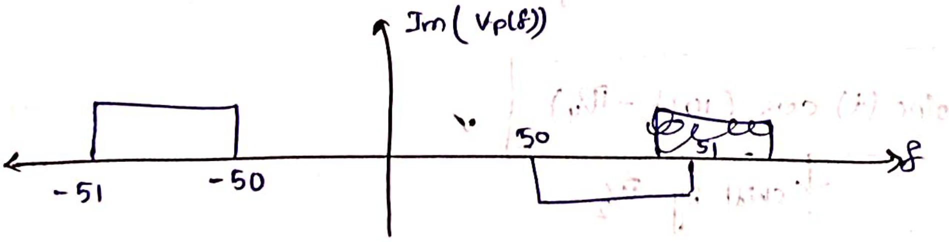
$$= \frac{1}{\sqrt{2}}\operatorname{sinc}(t)\left[\operatorname{sinc}(101\pi t) + \cos(101\pi t)\right]$$

$$m(t)\sin(2\pi f_c t) \xleftrightarrow{FT} \frac{1}{2j}\left[M(f-f_c) - M(f+f_c)\right]$$

$$m(t)\cos(2\pi f_c t) \xleftrightarrow{FT} \frac{1}{2}\left[M(f-f_c) + M(f+f_c)\right]$$

$$V_p(f) = \frac{1}{\sqrt{2}}\left[\frac{1}{2}\left[I_{[50,51]}(f) + I_{[-51,50]}\right] + \frac{1}{2j}\left[I_{[50,51]}(f) - I_{[-51,-50]}(f)\right]\right]$$





Bandwidth of $v_p(t)$: $51 - 50 = 1\,Hz$

(C) $\langle u_p, v_p\rangle = \frac{1}{2}\operatorname{Re}\{\langle u, v\rangle\}$

$$\langle u, v\rangle = \int_{-\infty}^{\infty} u(t)\cdot v^*(t)\cdot dt = \int_{-\infty}^{\infty} U(f)\, V^*(f)\, df$$

$$= \int_{-\infty}^{\infty} U(f)\cdot V^*(-f)\, df = \int_{0}^{1} \frac{1}{2} e^{+j\pi/4}\cdot df$$

$$= \int_0^1 \left(\frac{1}{2} e^{j\pi/4}\right) df = \frac{e^{j\pi/4}}{2}$$

$$\langle u_p, v_p\rangle = \frac{1}{2}\,\mathrm{Re}\left\{\frac{e^{j\pi/4}}{2}\right\}$$

$$= \cancel{\tfrac{1}{4\sqrt{2}}}\ \frac{1}{4}\cos(\pi/4) = \frac{1}{4\sqrt{2}}$$

$$\boxed{\langle u_p, v_p\rangle = \frac{1}{4\sqrt{2}}}$$

(d)

$$y_p(t) = (u_p * v_p)(t)$$

$$y(t) = \frac{1}{2}\left(u(t) * v(t)\right)$$ → complex BaseBand #

$$Y(f) = \frac{1}{2}\,U(f)\cdot V(f)$$

$$Y(f) = \frac{1}{2}\left(\frac{1}{2} I_{[-1,1]}(f)\right)\left(I_{[0,1]}(f)\, e^{-j\pi/4}\right)$$

$$Y(f) = \frac{1}{4} e^{-j\pi/4}\, I_{[0,1]}(f)$$

$I_{[0,1]} \leftrightarrow \mathrm{sinc}(t)\cdot e^{j\pi t}$

shift by $\frac{1}{2}$

$2\pi\cdot\frac{1}{2} = \pi$

Time domain

$$y(t) = \frac{1}{4} e^{-j\pi/4}\,\mathrm{sinc}(t)\, e^{j\pi t}$$

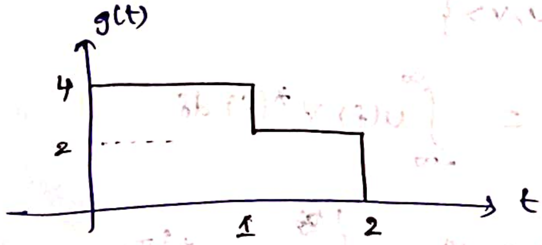
$$y_p(t) = \mathrm{Re}\left\{ y(t)\, e^{j100\pi t}\right\}$$

$$\boxed{y_p(t) = \frac{1}{4}\,\mathrm{sinc}(t)\cos(101t - \pi/4)}$$ ✓

shift by $\pi/2$

$$\boxed{\therefore\ y_p(t) = \frac{1}{4}\,\mathrm{sinc}(t)\cdot\sin(101\pi t + \pi/4)}$$ ✓ ☺

(2)



$$G(f) = \int_{-\infty}^{\infty} g(t)\, e^{-j2\pi f t}\, dt$$

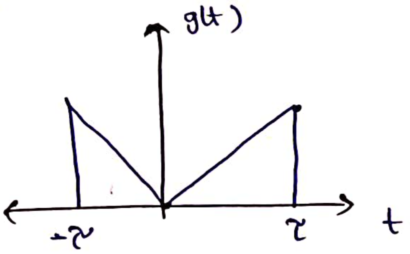
$$\therefore G(f) = 4\int_0^1 e^{-j2\pi f t}\, dt + \int_1^2 (2)\, e^{-j2\pi f t}\, dt$$

$$= \frac{4\left[e^{-j2\pi f} - 1\right]}{-j2\pi f} + \frac{2\left[e^{-j4\pi f} - e^{-j2\pi f}\right]}{-j2\pi f}$$

$$G(f) = 4e^{-j\pi f}\frac{\sin(\pi f)}{\pi f} + 2e^{-j3\pi f}\frac{\sin(\pi f)}{\pi f}$$

$$\boxed{\therefore G(f) = 4e^{-j\pi f}\,\text{sinc}(f) + 2e^{-j3\pi f}\,\text{sinc}(f)}$$

(b)

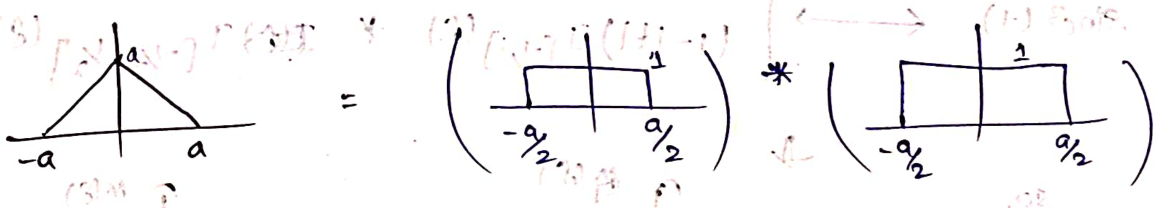


$$g(t) = \frac{1}{\tau}|t|\ \text{rect}\left(\frac{t}{2\tau}\right)$$

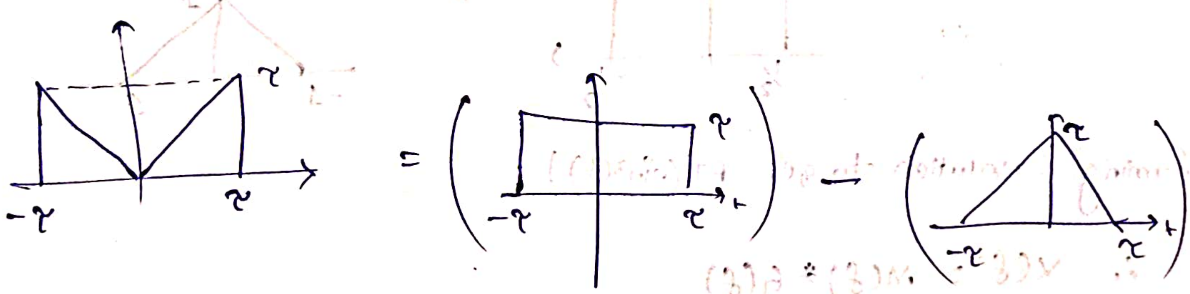
$$t = \left(1 - \left(1 - \left|\frac{t}{\tau}\right|\right)\right)\text{rect}\left(\frac{t}{2\tau}\right)$$

$$g(t) = \text{rect}\left(\frac{t}{2\tau}\right) - \left(1 - \frac{|t|}{\tau}\right)\text{rect}\left(\frac{t}{2\tau}\right)$$

We know that,

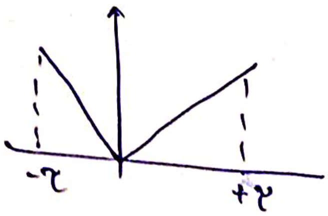


So,

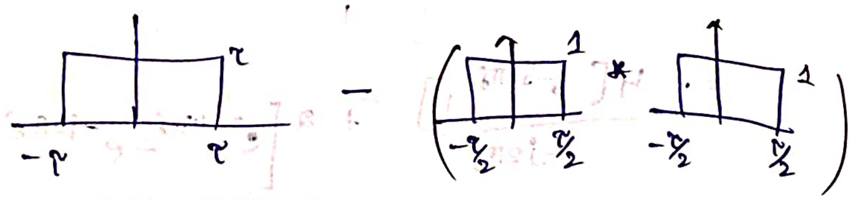


$$= |t|\ \text{rect}\left(\frac{t}{2\tau}\right)$$

so,



$= \tau . rect\left(\frac{t}{2\tau}\right) - \left(rect\left(\frac{t}{\tau}\right) * rect\left(\frac{t}{\tau}\right)\right)$

$=$ 

$$\tau\, G(f) = \tau\,(2\tau)\, sinc\,(f(2\tau)) - (\tau\, sinc\,(f\tau))^2$$

$$\tau . G(f) = 2\tau^2\, sinc\,(2f\tau) - \tau^2 sinc^2\,(f\tau)$$

$*$
$$\boxed{\therefore\; G(f) = 2\tau\, sinc(2\tau f) - \tau\, sinc^2(f\tau)}$$

③

(a) (i) $x(t) = sinc^3(t)$

We know that

$$sinc^2(t) \xleftrightarrow{FT} (1-|f|)\, I_{[-1,1]}(f)$$

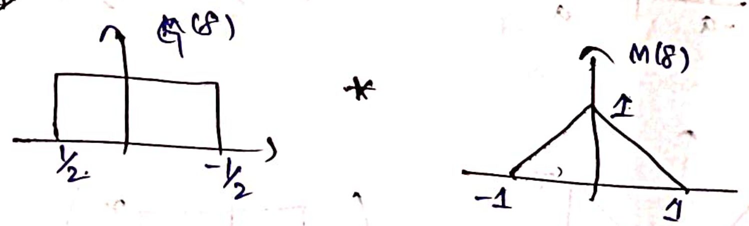
Also,

$$sinc(f) \xleftrightarrow{FT} I_{[-1/2,1/2]}(f) \qquad (\text{here } T=1)$$

$$sinc^3(t) \longleftrightarrow \left[(1-|t|)\, I_{[-1,1]}(f) * \cancel{I(f)}\; I_{[-1/2,1/2]}(f)\right]$$

$\Downarrow$

so,

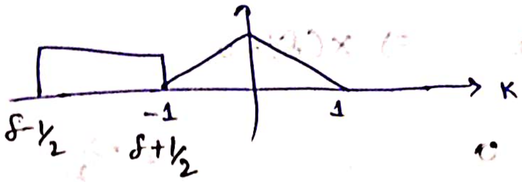


Performing convolution to get FT$(sinc^3(t))$,

$$\therefore\; X(f) = M(f) * G(f)$$

$$= \int_{-\infty}^{\infty} M(f)\, G(f-k)\, dk$$

Case-1: If $f < -3/2$



$$f + \tfrac{1}{2} < -1$$

$$\Rightarrow f < -3/2 \qquad x(f) = 0$$

Case-2: $f \in [-3/2, -1/2)$

$$x(f) = \int_{-1}^{f+1/2} (k+1)\, dk = \left[\frac{k^2}{2} + k\right]_{-1}^{f+1/2}$$

$$x(f) = \frac{(f+1/2)^2}{2} + (f+1/2) - \frac{1}{2} - 1$$

$$x(f) = \frac{f^2}{2} + \frac{3f}{2} + \frac{9}{8}$$

Case-3: $-1/2 \le f < 1/2$

$$x(f) = \int_{-1/2+f}^{0} (k+1)\, dk + \int_{0}^{f+1/2} (-k+1)\, dk$$

$$= \left[\frac{k^2}{2} + k\right]_{f-1/2}^{0} + \left[k - \frac{k^2}{2}\right]_{0}^{f+1/2}$$

$$= 0 - \left[\frac{(f-0.5)^2}{2} + (f - \tfrac{1}{2})\right] + (f+0.5) - \frac{(f+0.5)^2}{2}$$

$$= 1 - \frac{1}{2}\left((f+0.5)^2 + (f-0.5)^2\right)$$

$$= 1 - \left(f^2 + \frac{1}{4}\right) = 0.75 - f^2$$

Case-IV: $1/2 \le f < 3/2$

$$x(f) = \int_{f-1/2}^{1} (1-k)\, dk = \left[k - \frac{k^2}{2}\right]_{f-1/2}^{1}$$

$$= \frac{1}{2} - \left[(f - \tfrac{1}{2}) - \frac{(f-\tfrac{1}{2})^2}{2}\right]$$

$$= \frac{1}{2} - (f - \tfrac{1}{2}) + \frac{(f - \tfrac{1}{2})^2}{2}$$

$$= \frac{f^2}{2} - \frac{3f}{2} + \frac{9}{8}$$

Case-V: if $f \geq 3/2 \Rightarrow X(f) = 0$

$$X(f) = \begin{cases} 0 & f \leq -3/2 \\ \frac{f^2}{2} + \frac{3f}{2} + \frac{9}{8} & f \in [-3/2, -1/2] \\ 0.75 - f^2 & f \in [-1/2, 1/2] \\ \frac{f^2}{2} - \frac{3f}{2} + 9/8 & f \in [1/2, 3/2] \\ 0 & f \geq 3/2 \end{cases}$$

(3) (a)

(ii) $x(t) = t \cdot e^{-|\alpha| t} \cos(\beta t)\, u(t)$

$$e^{-|\alpha| t} u(t) \xleftrightarrow{FT} \frac{1}{j2\pi f + |\alpha|} = \frac{1}{j\omega + |\alpha|}$$

$e^{-at} u(t) \longleftrightarrow \frac{1}{s+a}$

$$\int_0^\infty e^{-|\alpha| t} \cdot e^{-j2\pi f t} = \int_0^\infty e^{-(j2\pi f + |\alpha|)t} = \frac{1}{j2\pi f + |\alpha|}$$

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t}\, dt$$

↓ diff wrt 'f' on both sides

$$\frac{d(X(f))}{df} = \int_{-\infty}^{\infty} (-j2\pi t)\, x(t)\, e^{-j2\pi f t}\, dt$$

$$t\, x(t) \longleftrightarrow \frac{dX(f)}{df} \cdot \left(\frac{1}{-j2\pi}\right)$$

$$\frac{-1}{j2\pi} \frac{d(X(f))}{df} = \int_{-\infty}^{\infty} (t \cdot x(t))\, e^{-j2\pi f t} \cdot dt$$

so

so,

$$\underbrace{t \cdot e^{-|\alpha| t} \cdot u(t)}_{k(t)} \longleftrightarrow \overbrace{\frac{(-1)}{(j2\pi f + |\alpha|)^2} \cdot \frac{j2\pi}{(-j2\pi)}}^{K(f)}$$

so,

$$t \cdot e^{-|\alpha| t} u(t) \cos(\beta t) \xleftrightarrow{FT} \frac{1}{2}\left[K\left(f - \frac{\beta}{2\pi}\right) + K\left(f + \frac{\beta}{2\pi}\right)\right]$$

$$\therefore\ t\cdot e^{-|\alpha| t}\cdot u(t)\cos(\beta t) \overset{FT}{\longleftrightarrow} \frac{1}{2}\left[\frac{1}{j2\pi\left(f-\frac{\beta}{2\pi}\right)+|\alpha|} + \frac{1}{j2\pi\left(f+\frac{\beta}{2\pi}\right)+|\alpha|}\right]$$

(3) (b) (ii) $\int_0^\infty e^{-\alpha t}\cos(\beta t)\,dt$

$$\int_{-\infty}^{\infty} x(t)\, e^{-j2\pi f t}\,dt = X(f)$$

We know

$\rightarrow$ ~~$f\, X(f)$~~

$$\Rightarrow \int_{-\infty}^{\infty} x(t)\,dt = X(0)$$

$\downarrow$ dc

$$X(0) = \int_{-\infty}^{\infty} e^{-\alpha t} u(t)\cos(\beta t)\,dt$$

$$x(t) = e^{-\alpha t} u(t)\cos(\beta t)$$

$$X(f) = \frac{1}{2}\left[\frac{1}{\left(j2\pi\left(f-\frac{\beta}{2\pi}\right)+\alpha\right)} + \frac{1}{\left(j2\pi\left(f+\frac{\beta}{2\pi}\right)+\alpha\right)}\right]$$

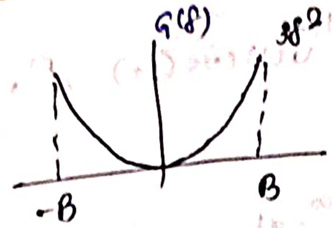
$$X(0) = \frac{1}{2}\left[\frac{1}{-j\beta+\alpha} + \frac{1}{j\beta+\alpha}\right]$$

$$= \frac{1}{2}\left[\frac{1}{\alpha-j\beta} + \frac{1}{\alpha+j\beta}\right]$$

$$= \frac{\alpha}{\alpha^2+\beta^2}$$

$$\therefore \int_0^\infty e^{-\alpha t}\cos(\beta t)\,dt = \frac{\alpha}{\alpha^2+\beta^2}$$

Q5: IFT of spectrum



(a) $G(f) = \begin{cases} 0 & \text{else} \\ 3f^2 & -B < f < B \end{cases}$

$$g(t) = \int_{-\infty}^{\infty} G(f)\, e^{j2\pi f t}\, df$$

$$\frac{d(g(t))}{dt} = \int_{-\infty}^{\infty} (j2\pi f)\, G(f)\, e^{j2\pi f t}\, df$$

$$\frac{d^2(g(t))}{dt^2} = \int_{-\infty}^{\infty} (j2\pi f)^2\, G(f)\, e^{j2\pi f t}\, df$$

$$\frac{3}{(j2\pi)^2}\, \frac{d^2(g(t))}{d^2t} = \int_{-\infty}^{\infty} 3f^2 \cdot G(f)\, e^{j2\pi f t}\, df$$

Let $G(f) = \text{rect}\left(f/2B\right)$

$$g(t) = 2B \sin(2Bt)$$

$$x(t) = \frac{3}{(j2\pi)^2}\, \frac{d^2 g(t)}{dt^2}$$

$$\boxed{x(t) = \frac{-3}{4\pi^2} \cdot \frac{d^2}{dt^2}\left(2B \sin(2Bt)\right)}$$

$$\cancel{x(t) = -\frac{3\times(2B)}{4\pi^2}\left(\frac{d}{dt}\left(\frac{d}{dt}\left(\frac{\sin(2\pi Bt)}{2\pi Bt}\right)\right)\right)}$$

$$\cancel{= -\frac{6B}{4\pi^2}\left(\frac{1}{2\pi B}\right)\left(\frac{d}{dt}\right)}$$

M-2:

$$g(t) = \int_{-B}^{B} 3f^2\, e^{j2\pi f t}\, dt$$

$$= 3\left[f^2 \frac{e^{j2\pi f t}}{(j2\pi t)} + \frac{2f\, e^{j2\pi f t}}{(2\pi t)^2} - \frac{2e^{j2\pi f t}}{j(2\pi t)^3} \right]_{-B}^{B}$$

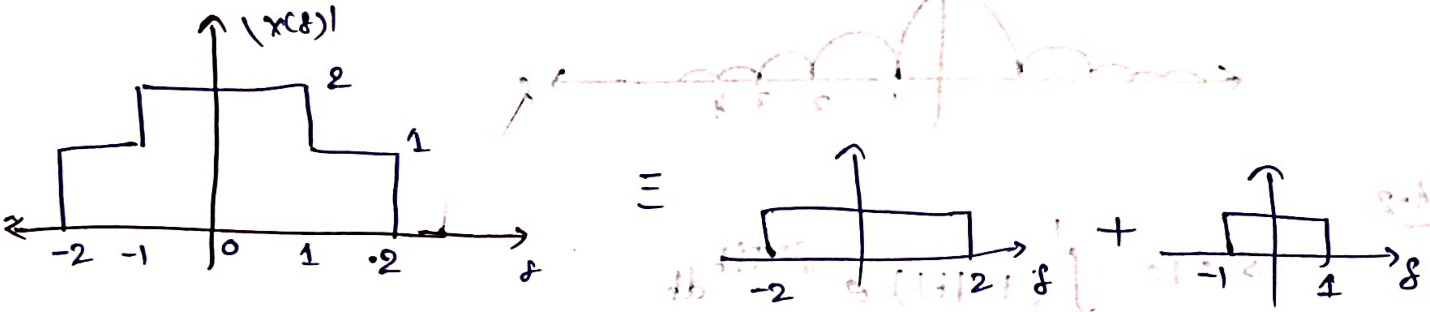
f^2	$e^{j2\pi f t}$
$-2f$	$\frac{e^{j2\pi f t}}{(j2\pi t)}$
$+2$	$\frac{e^{j2\pi f t}}{(j2\pi t)^2}$
-0	$\frac{e^{j2\pi f t}}{(j2\pi t)^3}$

$$g(t) = 3\left[\beta^2 \frac{\sin(2\pi Bt)}{\pi t} + B\frac{\cos(2\pi Bt)}{(\pi t)^2} - 4\frac{\sin(2\pi Bt)}{(2\pi t)^3}\right]$$

$$= 3\left[2\beta^3 \operatorname{sinc}(2\beta t) + \beta\frac{\cos(2\pi\beta t)}{(\pi t)^2} - \beta\frac{\operatorname{sinc}(2\beta t)}{(\pi t)^2}\right]$$

$$= \frac{-3}{(2\pi)^2}\frac{d^2}{dt^2}\left(2B \operatorname{sinc}(2Bt)\right) \quad // \qquad \because \operatorname{sinc}(2Bt) = \frac{\sin(2\pi Bt)}{2\pi\beta t}.$$

(b)



$$X(f) = \operatorname{rect}(f/4) + \operatorname{rect}(f/2)$$

$$\Downarrow \text{IFT}$$

$$\boxed{x(t) = 4\operatorname{sinc}(4t) + 2\operatorname{sinc}(2t)}$$

$$x(t) = \int_{-2}^{-1} e^{j2\pi ft}\,df + \int_{-1}^{1} 2e^{j2\pi ft}\,df + \int_{1}^{2} e^{j2\pi ft}\,df$$

$$x(t) = \frac{\left[e^{j2\pi ft}\right]_{-2}^{-1}}{j2\pi t} + 2\frac{\left[e^{j2\pi ft}\right]_{-1}^{1}}{j2\pi t} + \frac{\left[e^{j2\pi ft}\right]_{1}^{2}}{j2\pi t}$$

$$x(t) = 2\operatorname{sinc}(2t) + 4\operatorname{sinc}(4t)$$

$$x(t) = 2\frac{\sin(2\pi t)}{2\pi t} + 4\frac{\sin(4\pi t)}{4\pi t}$$

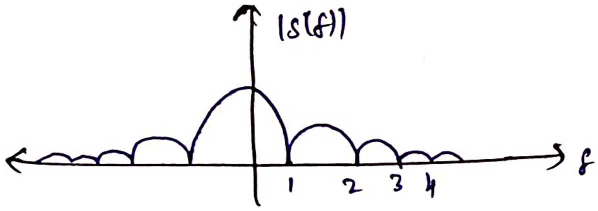
⑥ (a) $s(t) = (1-|t|)\, I_{[-1,1]}(t)$

$$= I_{[-1/2,1/2]}(t) * I_{[-1/2,1/2]}(t)$$

$\Downarrow$

$S(f) = \cancel{\text{rect}}\ \text{sinc}(f)\cdot\text{sinc}(f)$

$S(f) = \text{sinc}^2(f)$



M-2

$$S'(f) = \int_{-1}^{1} (1-|t|)\, e^{-j2\pi f t}\, dt$$

$$= \int_{-1}^{1} e^{-j2\pi f t}\, dt - \left[\int_0^1 t\, e^{-j2\pi f t} + \int_0^1 |-t|\, e^{j2\pi f t}\, dt\right]$$

$$= \int_{-1}^{1} e^{-j2\pi f t}\, dt - \left[\int_0^1 t\cdot 2\cos(2\pi f t)\, dt\right]$$

$$= \frac{\sin(2\pi f)}{\pi f} - \left[2\left[\frac{t\sin(2\pi f t)}{2\pi f} + \frac{2\cos(2\pi f t)}{(2\pi f)^2}\right]_0^1\right]$$

$$= \cancel{\frac{\sin(2\pi f)}{\pi f}} - \left(\cancel{\frac{\sin(2\pi f)}{\pi f}} + \frac{2\cos(2\pi f)}{(2\pi f)^2} - \frac{2}{(2\pi f)^2}\right)$$

$$= 2\left(\frac{1-\cos(2\pi f)}{(2\pi f)^2}\right) \qquad 1-\cos 2\theta = 2\sin^2(\pi f)$$

$$= \frac{\cancel{4}\sin^2(\pi f)}{\cancel{4}(\pi f)^2} = \text{sinc}^2(f)$$

6

(6)(b) $\int_{-B}^{B} |S(f)|^2 df = 0.99 \int_{-\infty}^{\infty} |S(f)|^2 df = 0.99 \int_{-\infty}^{\infty} |s(t)|^2 dt$

Parsevals eqn

on finding this from the below code of python

$B = 649\ Hz$ (one sided)

(4)(a) $s(t) = I_{[-1,1]}(t) \cos(400\pi t)$

$= Re\{ I_{[-1,1]}(t)\, e^{j400\pi t} \}$ $y_p(t) = Re\{u(t) e^{j2\pi f_c t}\}$

$= Re\{ \underbrace{I_{[-1,1]}\, e^{-j\pi t}}_{\text{complex envelope}} \cdot e^{j401\pi t} \}$

$s'(t) = I_{[-1,1]}(t)\, e^{-j\pi t} = s_c'(t) + j s_s'(t)$

$s(t) \longrightarrow \otimes \longrightarrow s_c'(t) = Re(s'(t))$

$\uparrow$

$2\cos(401\pi t)$

$= I_{[-1,1]}(t) \cos(\pi t)$

M-2

$2 s(t) \cos(401\pi t)$

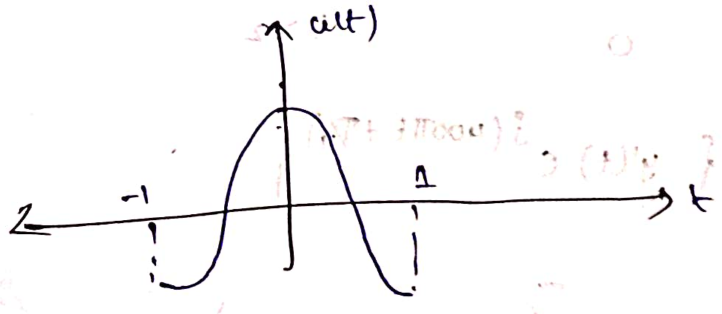
$= 2 I_{[-1,1]}(t) \cdot \cos(400\pi t) \cos(401\pi t)$

$2\cos A \cos B = \cos(A-B) + \cos(A+B)$

$= 2 I_{[-1,1]}(t) [\cos(\pi t) + \cos(801\pi t)]$

$\downarrow$ LPF

$= I_{[-1,1]}(t) \cos(\pi t) = u(t)$



(Q(b) $h(t) = I_{[0,1]}(t)\sin(400\pi t + \pi/4)$

$$= \mathrm{Re}\left\{ \underbrace{I_{[0,1]}(t)\ (-j)}_{h'(t)}\ e^{j(400\pi t + \pi/4)} \right\}$$

We know that,

$$y'(t) = \frac{1}{2}\left(s'(t) * h'(t)\right)$$

Here $y'(t) \rightarrow$ Complex envelope of $y(t)$

$$\therefore\ y'(t) = \frac{1}{2}\left(\underbrace{I_{[-1,1]}(t)\, e^{-j\pi/4}}_{s'(t)} * \underbrace{I_{[0,1]}(t)\ (-j)}_{h'(t)} \right)$$

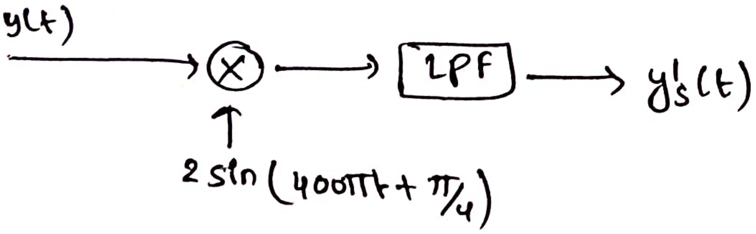
$$= \frac{-j e^{-j\pi/4}}{2}\left(I_{[-1,1]}(t) * I_{[0,1]}(t) \right)$$

$\Rightarrow I_{[-1,1]}(t) * I_{[0,1]}(t)$

$$= \begin{cases} 0 & ; t < -1 \\ \int_{-1}^{t} dx = t+1 & ; t \in [-1,0) \\ 1 & ; t \in [0,1) \\ \int_{t-1}^{1} dx = 2-t & ; t \in [1,2] \\ 0 & ; t > 2 \end{cases}$$

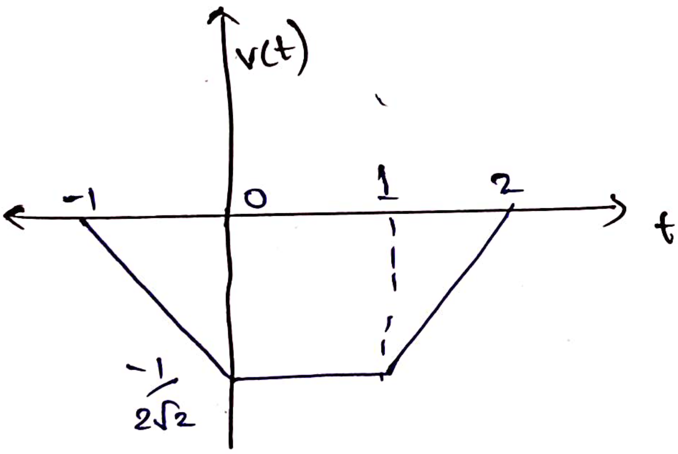
$$\therefore\ y'(t) = \begin{cases} 0 & t < -1 \\ -(t+1)e^{-j\pi/4}\left(\frac{j}{2}\right) & t \in [-1,0) \\ -\frac{e^{-j\pi/4} \times j}{2} & t \in [0,1] \\ -\frac{(2-t)\, e^{-j\pi/4} \cdot j}{2} & t \in [1,2] \\ 0 & t \geq 2 \end{cases}$$

$$y(t) = \mathrm{Re}\left\{ y'(t)\, e^{j(400\pi t + \pi/4)} \right\}$$



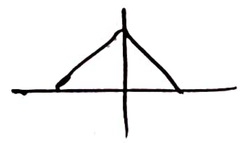
$$v(t) = \text{Imag}(y(t))$$

$$= \begin{cases} 0 & ; t < (-1) \\ -\dfrac{(t+1)}{2\sqrt{2}} & ; t \in [-1, 0) \\ -\dfrac{1}{2\sqrt{2}} & ; t \in [0, 1] \\ -\dfrac{(2-t)}{2\sqrt{2}} & ; t \in [1, 2] \\ 0 & ; t \geq 2 \end{cases}$$

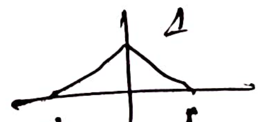


(3)(b)(i)

$$\int_0^\infty e^{-\alpha t}\,\mathrm{sinc}^2(t)\,dt = \int_{-\infty}^{\infty} e^{-\alpha t}.\,u(t)\,\mathrm{sinc}^2(t)\,dt$$



$$= \int_{-\infty}^{\infty} \frac{1}{\alpha + j2\pi f}\, M(f)\, df$$

$M(f) =$ 

$$= \int_{-1}^{0} \frac{f+1}{\alpha + j2\pi f}\, df + \int_{0}^{1} \frac{-f+1}{\alpha + j2\pi f}\, df$$

$$= \int \frac{x}{a+bx}\, dx = \frac{x}{b} - \frac{a}{b^2} \ln(a+bx)$$

$$= \int_0^\infty e^{-\alpha t}\,\mathrm{sinc}^2(t)\,dt$$

$$= \left[\frac{f}{j2\pi} + \frac{\alpha}{4\pi^2} \ln(\alpha + j2\pi f) \right]_{-1}^{0}$$

$$- \left[\frac{f}{j2\pi} + \frac{\alpha}{4\pi^2} \ln(\alpha + j2\pi f) \right]_{0}^{1} + \left(\frac{1}{j2\pi f} \ln(\alpha + j2\pi f) \right)_{0}^{1}$$

$$= \frac{1}{\pi} \tan^{-1}\left(\frac{2\pi}{\alpha}\right) + \frac{\alpha}{2\pi^2} \ln\left(\frac{\alpha}{\sqrt{\alpha^2 + 4\pi^2}} \right)$$